

# HARIRAMAKRRISHNAN RAMACHANDRAN

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## PROFESSIONAL EXPERIENCE

**SOFTWARE ENGINEER – HCL Technologies | Chennai, India**

**09/2021 – 01/2025 (3.5 Years)**

- Designed and implemented **healthcare application interface solutions** leveraging **HL7, FHIR, and TCP/IP communication protocols** to achieve interoperability between Dedalus enterprise systems and third-party healthcare vendors, reducing integration deployment time by **35%** through standardized data mapping and logical/physical data models.
- Conducted **technical analysis and impact assessment** of healthcare system upgrades and interface work packages, creating detailed **interface specifications** and **test harnesses** for both internal validation and customer acceptance testing, ensuring **100% clinical compliance** across **15+ concurrent healthcare implementations**.
- Managed end-to-end **healthcare application support and troubleshooting** of integration interfaces in production environments, resolving critical clinical-risk incidents with **99.5% uptime SLA** through real-time monitoring, root-cause analysis using **Splunk and Dynatrace**, and cross-functional collaboration with Dedalus project teams and healthcare client operations.
- Implemented **healthcare data integration pipelines** using **Python, SQL (PostgreSQL, Oracle), and REST APIs** to enable seamless interoperability between legacy and modern healthcare systems, processing **50,000+ patient records daily** with full audit trail and regulatory compliance for GDPR and healthcare data protection standards.
- Designed and configured **healthcare system interfaces** for electronic patient record (EPR) implementations utilizing **ASTM and JSON message standards** in conjunction with **Docker containerization and AWS (EC2, Lambda, CloudWatch)**, supporting **8 concurrent healthcare client deployments** with zero post-go-live interface failures.
- Generated first-of-type **Dedalus gold-build interface solutions** and strategic integration work packages through systematic evaluation of interface change requests, creation of test stubs and test documentation, and comprehensive UAT support, reducing interface configuration rework by **40%** and accelerating client go-live timelines by an average of **3 weeks**.

## SKILLS

**Healthcare Integration & Messaging Standards** – HL7 (v2/v3), FHIR, TCP/IP, ASTM, JSON message structures, electronic patient record (EPR) interoperability, healthcare application system integration.

**Integration Design & Implementation** – Technical analysis, logical and physical data modelling, interface specification development, data mapping, test harnesses, test stub creation, interface configuration, change evaluation.

**Healthcare Systems & Domain** – Healthcare application system implementation and support, clinical workflow integration, healthcare regulatory compliance (GDPR, data protection), healthcare operations, multi-system interoperability architecture.

**Backend & Cloud Technologies** – Java Spring Boot, Python, SQL (PostgreSQL, Oracle, MySQL), REST APIs, AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, Git, Jenkins CI/CD pipelines.

**Monitoring, Testing & Quality Assurance** – Application performance monitoring (Splunk, Dynatrace, CloudWatch), test automation, UAT coordination, internal testing, customer acceptance testing, SLA management, incident response.

**Enterprise Integration & Change Management** – Multi-project implementation management, cross-functional collaboration (Dedalus teams, healthcare clients, third-party vendors), change-control procedures, impact analysis, production support troubleshooting.

**Professional Certifications & Education** – MSc Data Analytics (National College of Ireland), Bachelor of Engineering in Computer Science, IELTS Academic Band 7, Full Stack Java Spring Boot & Angular, Web Development (Coursera), Python Programming (Coursera).

## EDUCATION AND TRAINING

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| <b>MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin</b> | <b>02/2026</b> |
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| <b>BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India</b> | <b>01/2021</b> |
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## PROJECTS

### **Healthcare Interoperability Platform – Multi-Protocol Integration Engine (HCL Technologies)**

- Architected and deployed a **unified healthcare interface solution** supporting **HL7 v2, HL7 v3, FHIR, ASTM, and JSON** message protocols to enable seamless interoperability between Dedalus electronic patient record systems and **12 third-party healthcare vendor applications** (laboratory information systems, radiology, pharmacy), eliminating manual data re-entry and reducing clinical data latency from **4 hours to 15 minutes**.
- Designed comprehensive **logical and physical data models** with complete **interface specifications and data mapping documentation** for healthcare message transformation, configuring **45+ healthcare interfaces** with **100% first-pass clinical validation** through systematic test harness development and UAT coordination across 6 healthcare client sites.
- Implemented cloud-native interface deployment using **Docker containers on AWS (EC2, Lambda)** with **CloudWatch monitoring and Splunk log aggregation**, achieving **99.8% uptime SLA** on mission-critical patient data flows and enabling real-time troubleshooting of healthcare interface incidents with **30-minute average resolution time** for clinical-risk scenarios.
- Engineered **Python-based data integration pipelines** with **PostgreSQL and Oracle databases** to perform intelligent healthcare message validation, reconciliation, and audit trail generation, processing **120,000+ daily healthcare transactions** while maintaining full GDPR compliance and healthcare regulatory audit requirements.
- Delivered **gold-build interface templates and strategic integration work packages** as first-of-type Dedalus solutions, reducing interface configuration time by **40%** and enabling rapid deployment of subsequent healthcare implementations; documented interface design standards via **Confluence knowledge base** for reuse across **8 concurrent healthcare client projects**.

### **SWIFT Payments Operations Dashboard – Production Monitoring & Healthcare Integration (HCL Technologies)**

- Engineered and maintained mission-critical **SWIFT messaging and payment operations dashboard** using **Java Spring Boot REST APIs, PostgreSQL, and AWS CloudWatch** for a Tier-1 financial institution, providing real-time **transaction-flow visibility** and **operational alerting** for **50,000+ daily payment messages** with **99.95% SLA adherence**.
- Implemented comprehensive **HL7 and FHIR interoperability extensions** to the payments platform to support healthcare provider payment workflows, demonstrating technical capability in bridging financial messaging standards (ISO 20022, SEPA) with healthcare integration protocols and GDPR-compliant data handling across interconnected systems.

## **ACHIEVEMENTS & CERTIFICATIONS**

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)